

change log

Added / Updated 03.25 - 04.08

- Cart improvements:
Display of cart summary (item count).
- Architecture improvements:
Fixed timezone for PHP and MySQL, ensuring time-based features (login locks) work correctly.
- Categories system:
 - Products now support categories.
 - Admin can create, update, and delete categories.
 - Each product can be assigned to one or multiple categories (multi-select)
 - Added category filters on the product listing page(s) to basic user and more advanced filters to admin (checkbox list on the left side).
 - Visibility Control:**(admin) Added an "Active/Inactive" toggle for products. Administrators can now hide products from the users.
- Safe category deletion logic:
 - Implemented handling of relationships when deleting categories:
 - Deleting a category does **not** delete the products within it.
 - Deleting a category does **not** break past orders that contain products from the deleted category.
- Admin panel improvements:
Added filtering and pagination for product management.
Filters include:
 - Category
 - Search by product name
 - Product status (active/inactive)
- Pagination:
Implemented pagination for product lists (9 items per page and admin page included).
- Filters & pagination via GET:
 - All filters and pagination states are now managed via GET parameters,
 - http_build_query ensures that if a user switches pages, their selected filters (e.g., search + category) are preserved
- Input validation improvements:
 - Added validation for all user inputs (emails, required fields, data types). This ensures data integrity and prevents corrupt entries.

- Basic brute-force protection:
 - Implemented IP-based login protection.
 - After multiple failed login attempts (base 3), login is temporarily blocked for a period of time (base 5 minutes).
- **SQL Injection Prevention:**
 - All search and filters realized via **Prepared Statements**, ensuring that user-provided input is never executed as part of the SQL command.

Added / Updated 04.08 - 04.11

Added

- Introduced order status system for products: "Ordered", "Processing", "Completed" for more realistic control of orders flow.
- Added new "Orders" admin panel with filtering capabilities:
 - Filter by order type/status
 - Search by order_id and user_id
 - Search by user_id
- added detailed information about order while EDIT
- added DELITE order possibility
- added pagination for orders list (10 items per page using LIMIT and OFFSET features for flowless work with big ammount of orders (to not reach rum cup on server)).

Improved

- Enhanced admin workflow for managing orders and tracking them.

Added / Updated 04.26

Architecture & Security Hardening

- **Database Performance (N+1 Query Fix):**
 - An **Eager Loading** architecture was implemented to address the classic "N+1" query problem. Product and category data is now retrieved using two efficient batch queries instead of a separate query for each product, which helps avoid database bottlenecks and ensures scalability
- **Concurrency & Race Condition Fix:**
 - Resolved a critical e-commerce race condition in the checkout process.
 - The system now uses an atomic SQL query `UPDATE ... WHERE stock >= ?` and carefully checks the value of `rowCount()` to ensure that an item cannot be resold,

even under heavy load. This ensures absolute integrity of inventory data.

- **Dynamic Image Processing Service:**

- A reliable ImageService has been implemented for secure server-side image processing.
- **Secure Uploads:** Validates all uploaded files by MIME type and file size to prevent malicious file uploads.
- **Automatic Resizing:** Automatically creates multiple image versions on upload: a lightweight thumbnail for fast-loading product lists and a high-quality large version for product detail pages.

- **Google reCAPTCHA Integration:**

- Implemented Google reCAPTCHA v2 on the login and registration forms to prevent automated bot attacks.
- **Resilient Architecture:** The backend verification uses a **Graceful Degradation** pattern, preferring the cURL library but seamlessly falling back to native PHP streams to ensure functionality on any server environment.
- **Improved Developer Experience (DX):** The .env.example is configured with Google's official test keys, allowing other developers to set up and run the project locally without needing to generate their own API keys.